Hybrid Intrusion Detection Using Wireshark Packet Analysis and Behavioral Modeling

Week 01 & Week 02

Project Benchmark Timeline

Phase	Duration	Key Tasks	Tools / Methods	Deliverable
1. Foundations & Planning	Week 1	Define scope (intrusions, dataset, approach), choose tools	Wireshark, tshark, Python	System architecture, feature list
2. Packet Capture & Parsing	Weeks 2–3	Capture packets, extract features (IP, ports, protocol, frequency), clean dataset	Scapy, Pandas	Structured dataset (CSV/DataFrame)
3. Behavioral Modeling	Weeks 4–6	Build detection logic (rule-based or ML), define anomalies	Rule-based OR Isolation Forest, K-Means	Detection engine
4. Integration	Week 7	Automate pipeline (capture → parse → detect), process PCAP files	PCAP scripts	End-to-end prototype
5. Testing & Evaluation	Weeks 8–9	Simulate attacks, measure detection accuracy and false positives	Nmap	Performance metrics
6. Visualization (Optional)	Week 10	Build dashboard, visualize alerts and traffic patterns	Matplotlib, Streamlit	Dashboard / visual reports

Prototype Level	Estimated Time	Scope
Minimal	4–5 weeks	Rule-based, offline analysis
Standard	6–8 weeks	Basic ML + automation
Advanced	8–10+ weeks	Real-time detection + visualization